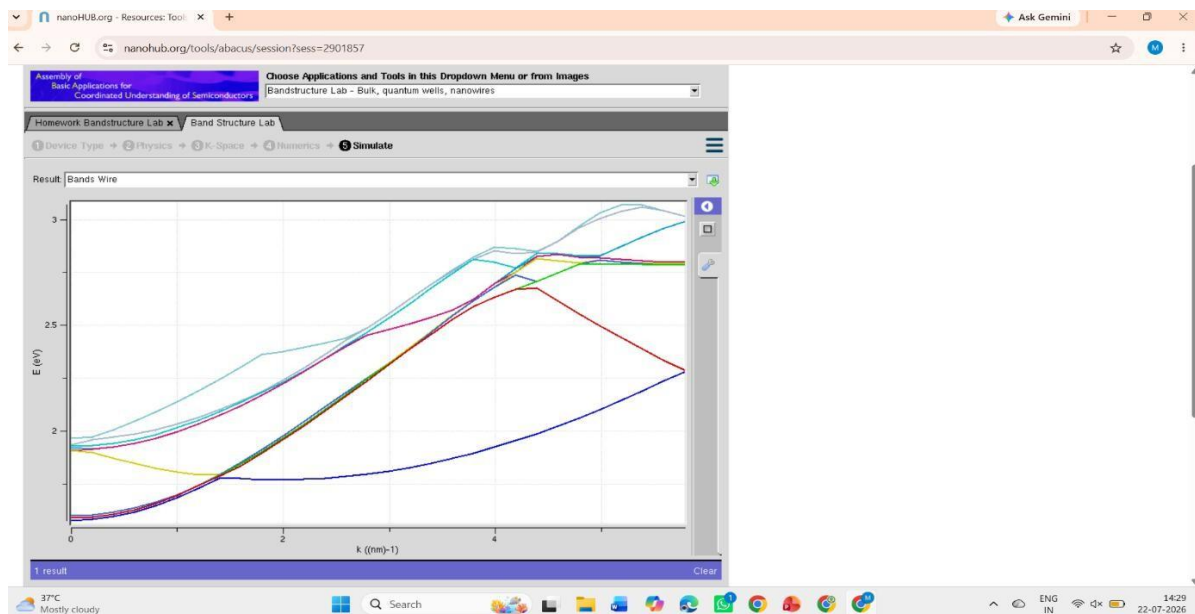


Exp. 5. Band Structure Simulation for Graphene and CNTs using Nano Hub

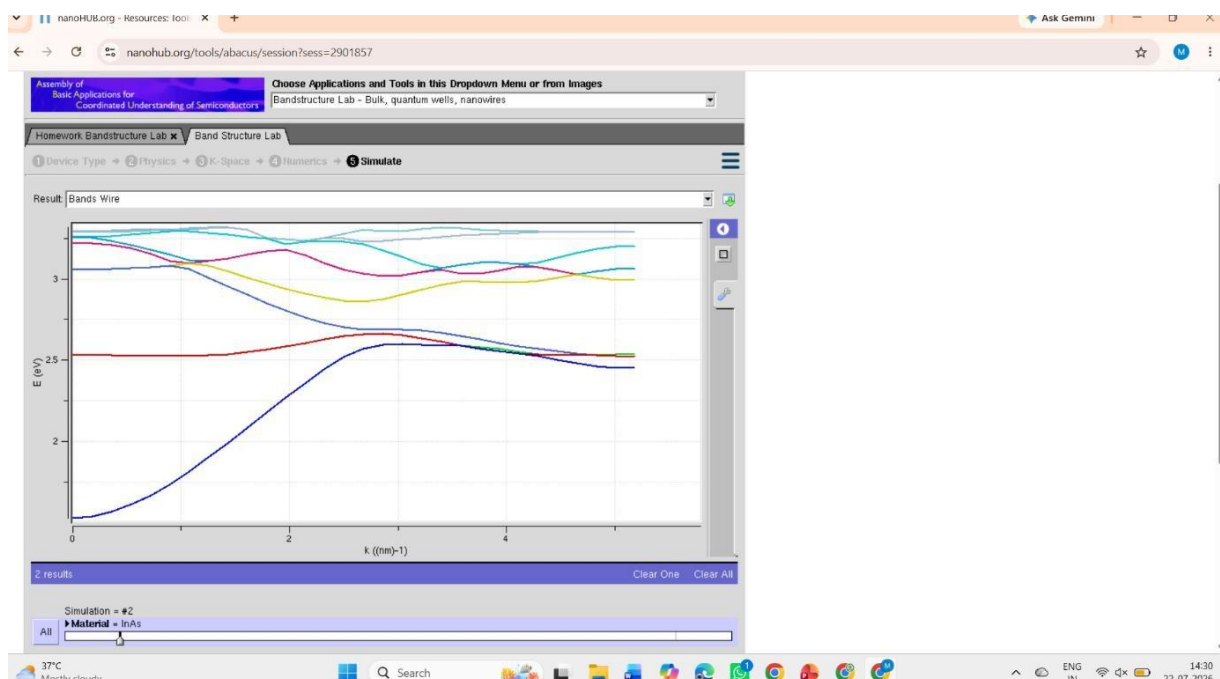
Aim: To simulate the band structure of nanowire for various materials.

Case Study 1: Simulate the band-structure of InAs circular nanowire



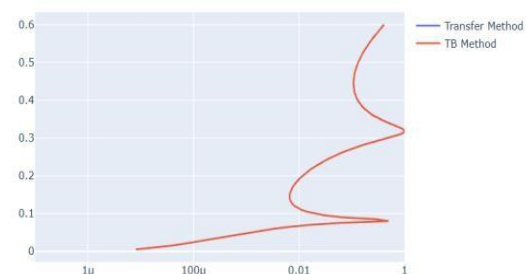
The graph shows the **energy band structure (E vs k)** of the semiconductor, where the energy of electrons changes with the wave vector. The different colored curves represent different energy bands, indicating the allowed energy levels for electron movement.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



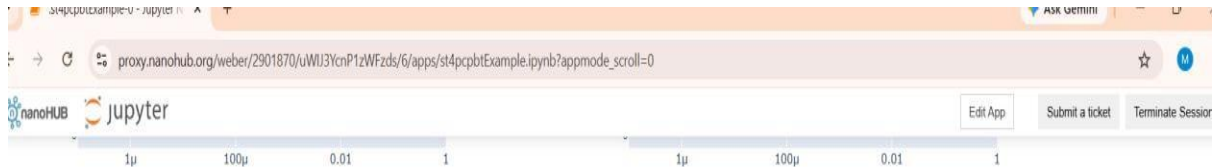
The graph compares **two band structure simulation results**, showing how the energy bands

Result: The band structure of **graphene and carbon nanotubes (CNTs)** was successfully simulated using **NanoHUB**. The results showed that **graphene has zero band gap**, while **CNTs can be metallic or semiconducting** depending on their structure. This experiment demonstrates the potential of graphene and CNTs for **high-speed nanoelectronic and semiconductor applications**.

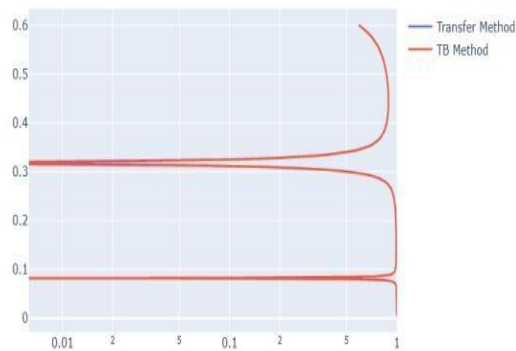


The transmission coefficient increases with electron energy, showing that electrons have a higher probability of passing through the potential barriers at higher energies. Both the Transfer Matrix and Tight-Binding methods produce nearly identical results, confirming the accuracy of the simulation.

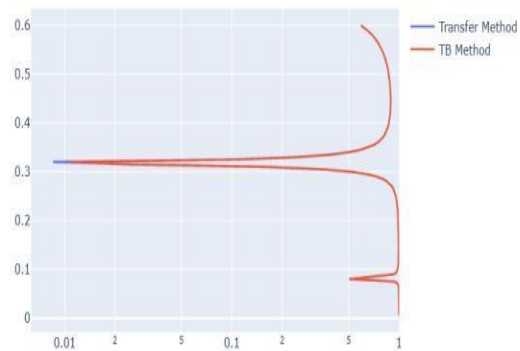
Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



Reflection Coefficient Vs Energy on refined grid

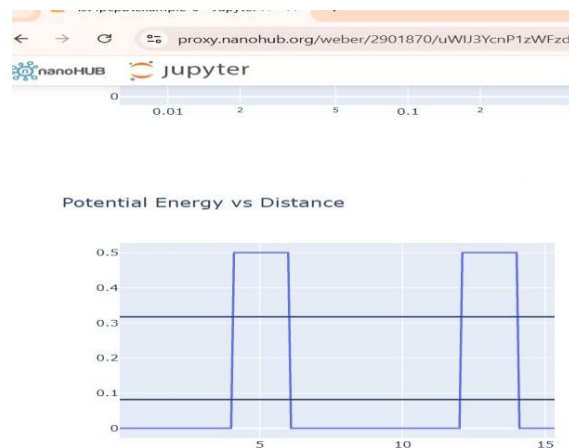


Reflection Coefficient Vs Energy on Homogeneous Grid

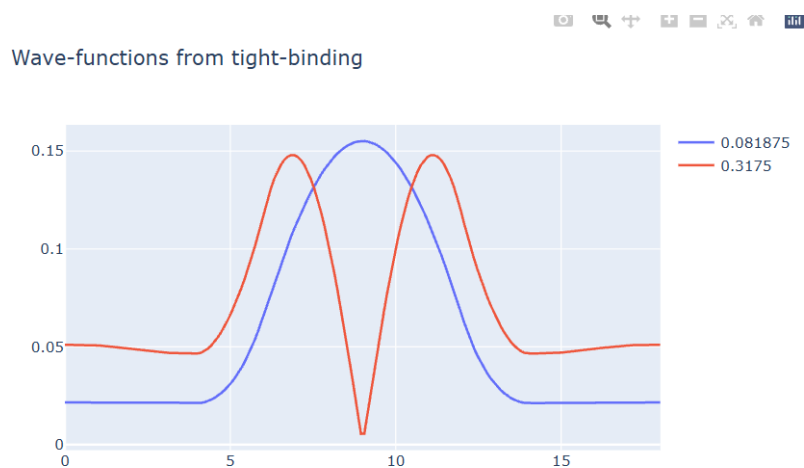


The reflection coefficient decreases as the transmission coefficient increases, indicating that fewer electrons are reflected at higher energies. The close overlap of both methods validates the consistency and reliability of the obtained results.

OUTPUT:



The graph shows a double potential barrier structure where the barrier height is 0.5 eV and the regions between barriers represent quantum wells. This potential profile controls the movement and tunneling behavior of electrons through the device.



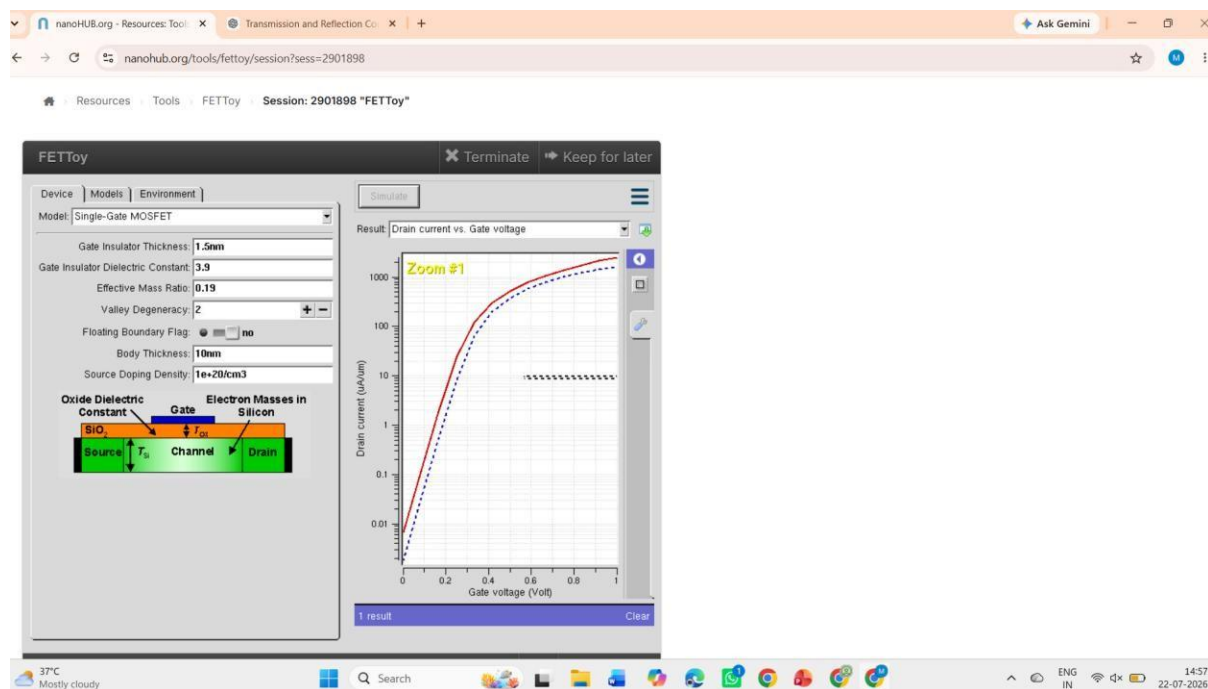
The wave functions are concentrated inside the quantum well, indicating a high probability of finding electrons in this region. The different energy states exhibit distinct wavefunction shapes, demonstrating quantum confinement and resonant behaviour.

Result: The quantum tunneling phenomenon was successfully demonstrated, showing that electrons can tunnel through a potential barrier even when their energy is lower than the barrier height. The results indicate that the tunneling probability decreases as the barrier height and width increase. This experiment confirms the principles of quantum mechanics and highlights the importance of quantum tunneling in nanoelectronic devices such as tunnel diodes and scanning tunneling microscopes (STM).

Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using Nano Hub

Case 1: Increase in V_G :

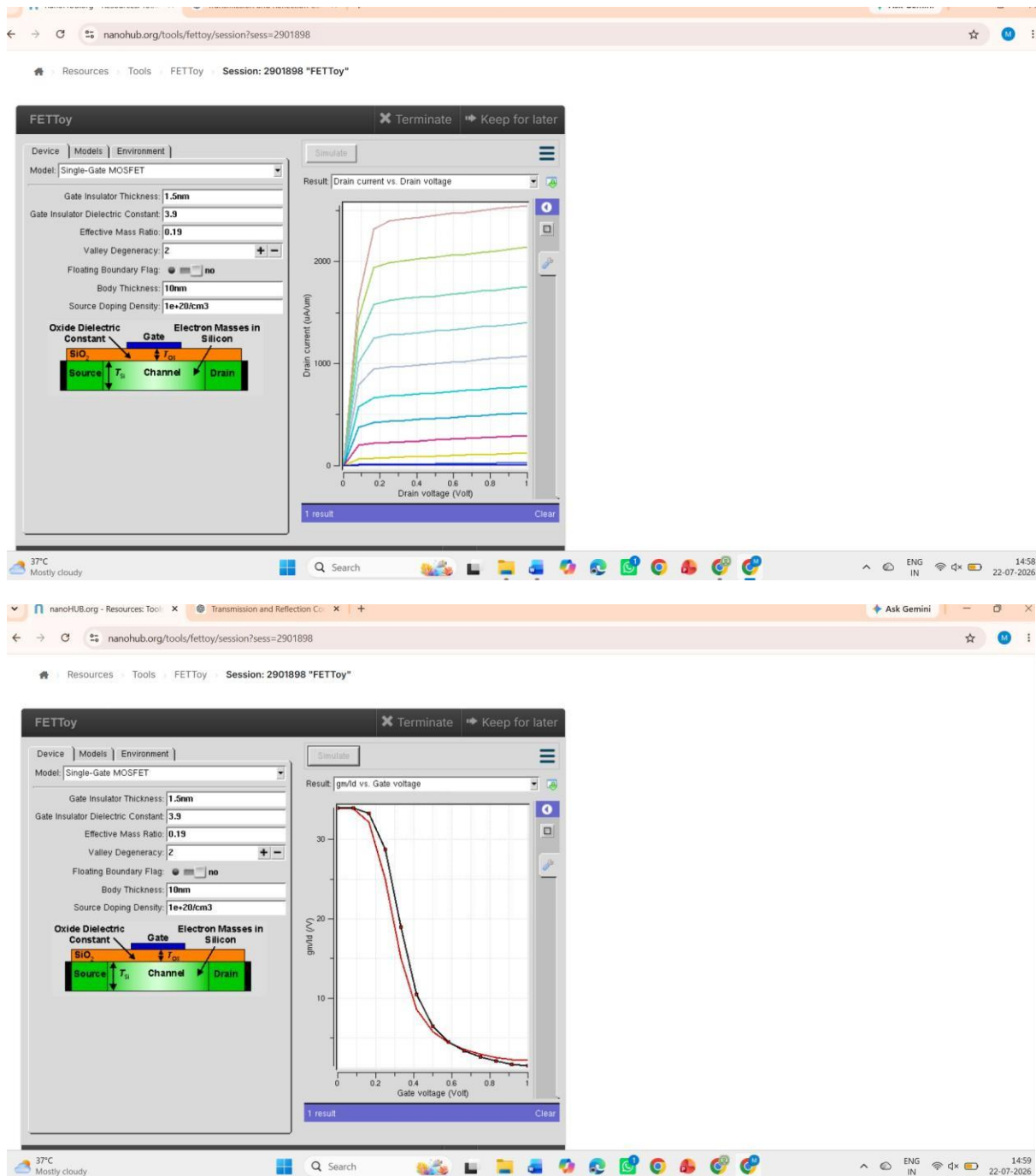
- **Observation:** As V_G increases, the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.



The graph shows that the drain current (I_D) increases rapidly with increasing drain voltage (V_D) and then reaches a saturation region. Higher gate voltages produce higher drain currents, demonstrating the normal operating characteristics of a Single-Gate MOSFET.

Case 2: Variation in V_D :

- **Observation:** Drain current (I_D) saturates for higher V_D at a fixed V_G :



The graph shows that the drain current (I_d) increases exponentially after the threshold voltage is reached as the gate voltage (V_g) increases. This confirms that the MOSFET turns ON and allows more current to flow with increasing gate voltage.

Results: The MOSFET scaling analysis and subthreshold swing calculation were successfully performed using NanoHUB. The results showed that reducing the device dimensions improves switching speed and integration density, while the subthreshold swing reflects the efficiency of transistor switching. This experiment demonstrates the importance of device scaling and subthreshold characteristics in designing high-performance, low-power MOSFETs for modern integrated circuits.